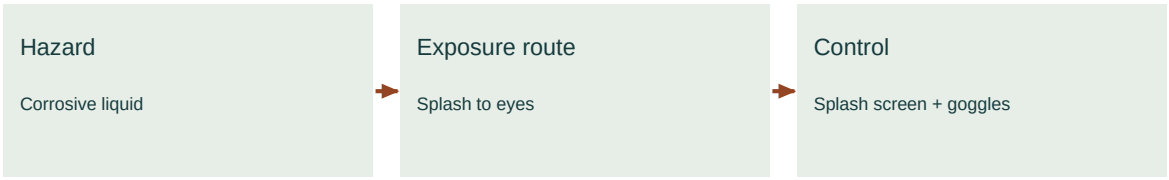
06

Instruments and Lab Safety

A field guide to understanding, working and checking.

Make the safety reasoning visible



Controls must match the actual procedure and local safety instructions.

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

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8 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Hazard, risk and prevention
2. Personal protective equipment and contamination
3. Exposure routes and toxic effects
4. Labels, SDS and chemical properties
5. Glassware, measurement and safe apparatus
6. Hoods, heating, cooling and pressure
7. Waste, housekeeping and emergency priorities
8. Biosafety and the control hierarchy

LESSON 01 / 8

Hazard, risk and prevention

After this lesson, you can

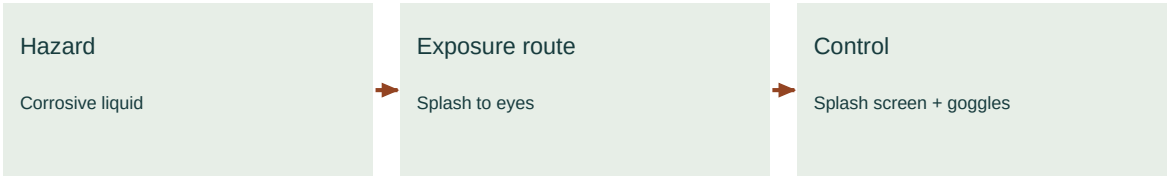
- Distinguish a hazard from the risk of harm in a particular situation.
- Choose a control that interrupts the actual exposure route.

Start with: Hazard means potential for harm; exposure means contact with the harmful agent.

THEORY

A hazard is a source of possible harm. Risk depends on exposure and the chance and severity of harm. Accident prevention is shared: prepare, follow approved procedures and report unsafe conditions. A familiar chemical or routine experiment can still present hazards.

Make the safety reasoning visible



Controls must match the actual procedure and local safety instructions.

Link the hazard to a specific exposure route and suitable controls. Eye protection is one layer, not a substitute for eliminating or reducing the hazard.

RULE

Good answer: hazard → exposure route → control

A method that earns marks

1. Name the harmful property and the activity creating exposure.
2. Identify who could be exposed, by which route and under what conditions.
3. Select a control that reduces the source or interrupts the route.
4. Explain the remaining need for the approved procedure and required personal protection.

WORKED EXAMPLE

A volatile harmful liquid on an open bench creates an inhalation hazard. Using the approved fume-hood procedure reduces exposure; goggles alone do not control vapour inhalation.

1. The example liquid is volatile and harmful, so vapour can leave an open container.
2. A person nearby may inhale that vapour; this connects the chemical hazard to an exposure route.
3. An approved fume-hood procedure controls vapour at its source when suitable for that substance and operation.
4. Goggles protect eyes from appropriate hazards but do not remove vapour from inhaled air.

Common mistake
Do not write only be careful; name a concrete control.

Exam habit: For a scenario answer write a linked chain: hazard, exposure route, suitable control and reason.

SELF-CHECK

Why is naming only a hazardous chemical insufficient to describe a risk scenario?

Answer: Risk also depends on how exposure can occur, how likely it is and how serious its consequences could be.

LESSON 02 / 8

Personal protective equipment and contamination

After this lesson, you can

- Trace how contaminated gloves transfer material to clean surfaces.
- Select PPE for the task and remove it without spreading contamination.

Start with: Contamination transfer; clean and contaminated work zones; task-specific protection.

THEORY

Splash goggles protect against chemical splashes; ordinary spectacles do not provide equivalent coverage. Gloves must be compatible with the chemical and inspected. No glove protects indefinitely. Contaminated gloves can carry chemicals to phones, door handles and notebooks.

RULE

Select PPE for the specific hazard and task

A method that earns marks

1. Identify the contaminant and which surfaces or body areas need protection.
2. Use PPE specified for the material and task rather than assuming one glove type protects against everything.
3. Keep contaminated gloves away from phones, door handles and other clean shared items.
4. Remove and dispose of PPE according to the approved procedure, then perform required hand hygiene.

WORKED EXAMPLE

A student touches a phone with contaminated gloves. The issue is transfer of contamination to a surface later touched without gloves. Remove gloves correctly before touching clean shared items.

1. The student glove is already contaminated before touching the phone.

2. Contact transfers material from glove to phone, creating a new contaminated surface.
3. Later bare-hand contact can move that material to skin, food or the face.
4. Removing gloves correctly before touching clean items breaks that transfer chain; the phone may still need the approved decontamination procedure.

Common mistake
More layers of unsuitable gloves do not establish chemical compatibility.

Exam habit: Explain what is transferred and to where; simply saying wear gloves does not solve a contamination scenario.

SELF-CHECK

Do gloves make it acceptable to touch your face during laboratory work?

Answer: No. Gloves can carry contamination directly to the face.

LESSON 03 / 8

Exposure routes and toxic effects

After this lesson, you can

- Identify inhalation, ingestion, skin or eye contact and injection exposures.
- Distinguish route of exposure from the timing of a toxic effect.

Start with: Contact with an agent can occur by several routes; visible vapour is not required.

THEORY

Chemicals can enter by inhalation, ingestion, skin or eye absorption, or injection through contaminated sharps. Acute exposure is short-term; chronic exposure is repeated or long-term. Effects can be local or systemic, immediate or delayed, reversible or irreversible.

RULE

Dose, duration, route and substance all influence harm

A method that earns marks

1. Read the activity and locate the point where the agent could enter or contact the body.
2. Name the route using the specific evidence in the scenario.

3. Distinguish a rapid acute effect from an effect associated with longer or repeated exposure.
4. Match prevention to the identified route and follow the material-specific procedure.

WORKED EXAMPLE

Eating in a laboratory can lead to ingestion of contamination. A contaminated needle can create injection exposure even without visible vapour.

1. Eating in the laboratory can move contamination from hands or surfaces to food.
2. Swallowing that material is ingestion, even when the contaminant is not visible.
3. A contaminated needle can breach the skin and introduce material directly: this is injection exposure.
4. The absence of visible vapour does not remove either risk because neither route depends on inhalation.

Common mistake
A lack of smell does not prove safety.

Exam habit: Name both the route and the evidence, for example ingestion because contaminated food is swallowed.

SELF-CHECK

A droplet splashes into an eye. Is the route inhalation?

Answer: No. It is eye or mucous-membrane contact.

LESSON 04 / 8

Labels, SDS and chemical properties

After this lesson, you can

- Use a label and SDS to connect chemical properties with controls.
- Distinguish corrosive and oxidizing hazards in an explanation.

Start with: Chemical identity; label symbols; an SDS gives substance-specific hazard and handling information.

THEORY

Read the identity and hazard information before use. Safety data sheets describe hazards, handling, storage, exposure controls, first aid and other substance-specific information. Flammable, oxidizing, corrosive and toxic refer to different hazards and can apply to one substance simultaneously.

RULE

$$\text{Identity} + \text{hazard} + \text{controls} + \text{incompatibilities}$$

A method that earns marks

1. Confirm the chemical identity from its label before selecting a procedure.
2. Read relevant SDS information for hazards, incompatibilities and required handling controls.
3. Explain how the named property causes harm in the scenario.
4. Use the approved storage or handling arrangement rather than inferring compatibility from appearance.

WORKED EXAMPLE

A corrosive liquid can damage tissue, while an oxidizer may intensify fire even when it is not itself a fuel. These require different control reasoning.

1. A corrosive liquid can damage living tissue or certain materials by contact.
2. An oxidizer can support or intensify combustion even when it is not itself the fuel.
3. The mechanisms differ, so the same general statement handle carefully does not explain either control.
4. A good answer links splash/contact controls to corrosivity and incompatibility or fire-prevention controls to oxidizing behaviour as specified by the SDS.

Common mistake
Do not infer that all colourless liquids are harmless or that one symbol captures every hazard.

Exam habit: Pair each property with its consequence; listing pictogram names alone gives less evidence of understanding.

SELF-CHECK

Does a chemical being nonflammable prove it cannot intensify a fire?

Answer: No. An oxidizer may intensify combustion without being a fuel.

LESSON 05 / 8

Glassware, measurement and safe apparatus

After this lesson, you can

- Select apparatus according to measurement purpose and safe operating limits.
- Explain why volumetric preparation and heating require different vessels.

Start with: Volume; calibration mark; damaged glassware can fail during use.

THEORY

Inspect glassware for chips and cracks and apparatus for damaged wiring before use. A beaker is useful for holding and mixing; a graduated cylinder gives more useful volume estimates; a volumetric pipette or flask is designed for a specified accurate volume. Read liquid levels at eye height using the appropriate meniscus.

RULE

Choose the instrument for required precision, not convenience

A method that earns marks

1. State whether the task requires approximate volume, accurate delivery or preparation to a final volume.
2. Choose the apparatus designed for that purpose and check its condition.
3. Follow the specified filling, meniscus and operating procedure.
4. Before heating, verify that the vessel and setup are approved for the process.

WORKED EXAMPLE

A volumetric flask prepares a solution to its marked final volume. It is not the appropriate vessel for directly heating a reaction.

1. The volumetric flask is calibrated to contain a stated final volume at its reference conditions.
2. The solution is brought to the mark as part of an approved preparation procedure.
3. Its main role is volume accuracy, not serving as a directly heated reaction vessel.
4. For heating a reaction, use the vessel and support specified by the laboratory method rather than substituting the flask.

Common mistake
Extra decimal places do not make an imprecise instrument accurate.

Exam habit: In apparatus questions state the purpose and limitation together: prepares final volume, unsuitable for directly heating the reaction.

SELF-CHECK

Does having a calibration mark mean a flask is suitable for direct heating?

Answer: No. Calibration and suitability for heating are different design requirements.

LESSON 06 / 8

Hoods, heating, cooling and pressure

After this lesson, you can

- Explain why heating a closed unsuitable vessel can create a pressure hazard.
- Recognize when an apparatus setup requires instructor review.

Start with: Heating can increase vapour production and gas pressure; equipment has specific operating limits.

THEORY

A fume hood controls airborne chemical exposure when used correctly. It is not a storage cabinet or a waste-evaporation device. Heating and cooling systems can introduce burns, pressure changes and material failure. Approved setups account for both temperature and pressure.

RULE

Control vapour, ignition, pressure and thermal contact separately

A method that earns marks

1. Identify heat source, volatile materials and any sealed volume.
2. Check the approved procedure and apparatus pressure and temperature limits.
3. Use the specified containment, heating and ventilation arrangements.
4. If the setup differs or becomes unsafe, stop and alert the responsible instructor instead of modifying it ad hoc.

WORKED EXAMPLE

A volatile solvent warmed in an unsuitable sealed container can build pressure. The correct response is to stop and consult the approved setup rather than improvise a vent.

1. Warming the volatile solvent increases vapour formation.
2. In an unsuitable sealed container, expanding gas and vapour can increase internal pressure.
3. The container or closure may fail because it was not designed for that operation.
4. The example therefore calls for stopping and consulting the approved setup; improvising a vent is not a validated correction.

Common mistake
A hood does not remove every fire, pressure or contact hazard.

Exam habit: Explain the mechanism, pressure buildup from heating, before giving the required stop-and-consult response.

SELF-CHECK

Does placing an unsuitable sealed heated vessel in a fume hood solve its pressure hazard?

Answer: No. A hood does not make a vessel pressure-rated or validate an unsafe heating setup.

LESSON 07 / 8

Waste, housekeeping and emergency priorities

After this lesson, you can

- Use identification and approved procedures to choose a waste or spill response.
- State safe priorities when a spill material is unknown.

Start with: Chemical incompatibility; designated waste streams; laboratory emergency instructions.

THEORY

Waste must follow the laboratory segregation and disposal procedure. Broken glass, sharps and chemical waste are different streams. Good housekeeping keeps exits and emergency equipment accessible. In an incident, alert others and follow the established response; do not improvise a hazardous cleanup.

RULE

Recognize → alert → follow trained response

A method that earns marks

1. Identify the material only if that can be done safely from available records or labels.
2. Keep others clear and alert the instructor or responsible responder.
3. Use the laboratory response plan for the identified hazard and your training level.
4. For routine waste, use the designated compatible container and labelling procedure.

WORKED EXAMPLE

An unknown spill is not an invitation to mix in a neutralizer. Alert the instructor, keep others clear and follow the spill response appropriate to the identified material.

1. An unknown spill has uncertain properties, so an assumed neutralizer may be incompatible.
2. Adding another chemical could cause heat, gas formation or another hazardous reaction.
3. Alerting the instructor and keeping people clear limits exposure while the material and response are assessed.
4. Cleanup then follows the approved spill procedure for the identified material and trained personnel.

Common mistake
Do not recommend that an untrained student fight every fire or clean every spill.

Exam habit: In an emergency scenario prioritize alerting and exposure prevention; do not invent a chemical cleanup recipe.

SELF-CHECK

Should an unknown laboratory liquid automatically be washed into a sink?

Answer: No. Its identity, compatibility and approved disposal route must be established.

LESSON 08 / 8

Biosafety and the control hierarchy

After this lesson, you can

- Classify controls within the hazard-control hierarchy.
- Distinguish engineering containment from training, work practices and PPE.

Start with: A biological hazard may be spread through aerosols, contact or sharps depending on the activity.

THEORY

Biosafety aims to prevent unintended exposure to biological agents and their release. The general control hierarchy prefers removing a hazard, then using a safer substitute, then engineering controls, administrative controls and work practices, and PPE. The suitable controls depend on the task: removing or replacing the biological material may not be possible. Training and procedures are administrative controls; suitable containment equipment is an engineering control. Safe work practices include hand hygiene and avoiding aerosol generation.

RULE

Eliminate → substitute → engineering → administrative/work practices → PPE

A method that earns marks

1. Identify the biological material and activity-specific exposure route.
2. Consider elimination or substitution where feasible before downstream controls.
3. Classify equipment containment as engineering, training and procedures as administrative or work-practice controls, and worn barriers as PPE.
4. Explain how the selected combination addresses the task and follows institutional biosafety procedures.

WORKED EXAMPLE

Training is administrative. A suitable biological safety cabinet is engineering containment. Gloves are PPE. A safe handling technique is a work practice.

1. Training changes how people work, so it is an administrative control.
2. A suitable biological safety cabinet provides engineering containment for its approved uses.
3. Gloves form a worn barrier and are PPE.

4. A safe handling technique is a work practice, generally grouped with administrative controls; the categories describe different protective mechanisms.

Common mistake
A device with airflow is not automatically suitable for every chemical or biological hazard.

Exam habit: Do not label every cabinet or hood as interchangeable containment; state suitable equipment for the defined biological task.

SELF-CHECK

What is the general hierarchy from source-focused to last-line protection?

Answer: Elimination, substitution, engineering controls, administrative controls and work practices, then PPE.

The control hierarchy in the correct order

The hierarchy ranks ways to control a hazard. A list of control types is not automatically in the preferred order.

- Eliminate: remove the hazard.
- Substitute: use a safer alternative after checking its risks.
- Engineering controls: separate people from the hazard using suitable equipment or containment.
- Administrative controls and work practices: use training, procedures and safe methods.
- PPE: use suitable worn protection. It adds protection but depends on correct selection and use.
- In a laboratory, eliminating or replacing the material may not be possible. The assessed task determines the suitable combination of controls.

Classify a suitable containment cabinet, safety training, gloves and correct handwashing.

1. The cabinet is an engineering control.
2. Training is administrative; gloves are PPE.
3. Handwashing is a safe work practice, often grouped with administrative controls.

Engineering; administrative; PPE; work practice, respectively. Engineering controls come before administrative controls and PPE in the general hierarchy.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Instruments and Lab Safety / Test A

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which best describes risk?

   A. The mass of any bottle

   B. The same as smell

   C. The label colour alone

   D. The chance and severity of harm under the exposure conditions

2. Which is appropriate before touching a clean door handle?

   A. Keep contaminated gloves on

   B. Wipe gloves on clothing

   C. Remove contaminated gloves correctly

   D. Smell the gloves

3. Eating with contaminated hands is mainly which exposure route?

   A. Inhalation

   B. Injection

   C. Noise exposure

   D. Ingestion

4. An oxidizer is especially associated with:

   A. Intensifying combustion

   B. Preventing all fires

   C. Measuring temperature

   D. Making all chemicals inert

Part II: Show your reasoning

1. A student uses an unfamiliar reagent without reading its safety information. Identify the problem and two appropriate preparatory actions.

2. Explain why ordinary prescription glasses and a lab coat alone are insufficient for a splash-risk task.

3. Classify inhaling vapour and a contaminated needle puncture by exposure route. Explain why no smell does not prove no hazard.

4. A bottle has both flammable and corrosive hazards. State two distinct risks and a control principle for each.

Instruments and Lab Safety / Test B

20 marks | Suggested time: 20 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which is designed to prepare one accurately specified solution volume?

   A. A beaker

   B. A watch glass

   C. A volumetric flask

   D. A test-tube rack

2. A fume hood should be used as:

   A. Permanent chemical storage

   B. A controlled work enclosure for appropriate vapour hazards

   C. A way to evaporate all waste

   D. A food cupboard

3. An unknown chemical spill should first prompt a student to:

   A. Alert the instructor and keep people clear

   B. Mix it with another chemical

   C. Ignore it

   D. Taste it

4. Which is an administrative control?

   A. Gloves

   B. Training and written procedures

   C. A sealed rotor

   D. Splash goggles

Part II: Show your reasoning

1. Name a suitable device to transfer an accurate fixed 10 mL volume and explain two good practices.

2. Give two reasons why heating a tightly sealed vessel containing a volatile liquid is hazardous.

3. A box blocks access to the eyewash station. Explain the risk and an appropriate preventive action.

4. Classify goggles, safety training, a suitable containment cabinet and correct handwashing.

Instruments and Lab Safety / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **D.** Risk depends on both the hazard and how people are exposed.
2. **C.** This prevents spreading contamination outside the work area.
3. **D.** Material can pass from the hands to food and the digestive system.
4. **A.** Oxidizers can support or intensify combustion.

Part II: Written solutions

1. Problem: unassessed chemical hazards. Actions: consult the reagent SDS and approved procedure; clarify required controls and waste handling with the instructor before use. One mark for hazard, one for exposure reasoning and one per specific action.
2. Prescription glasses do not seal against splashes; appropriate splash goggles are needed. A coat does not protect hands, so compatible gloves may be required by the assessed procedure. Award one mark per protection gap and suitable control.
3. Vapour: inhalation. Needle: injection. Some harmful substances are odourless, or odour detection is unreliable. One mark per route and two for the explanation.
4. Fire risk: control ignition sources using the approved setup. Tissue damage from splashes: prevent contact using appropriate containment and compatible protection. One mark per risk and per linked control.

Instruments and Lab Safety / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **C.** The calibration mark defines the specified contained volume under its calibration conditions.
2. **B.** It is an engineering control for relevant airborne exposures.
3. **A.** Unknown hazards require assessment and the established response.
4. **B.** Administrative controls organize how work is authorized and performed.

Part II: Written solutions

1. Use an appropriate 10 mL volumetric pipette with a pipette aid. Read the meniscus at eye level and follow the calibrated delivery procedure. Credit device, aid and two measurement practices.
2. Vapour generation can raise pressure, and thermal expansion can further stress the vessel. Rupture can release hot liquid or vapour. Credit two mechanisms and their resulting consequences.
3. It can delay emergency irrigation after exposure. Keep access clear and report the obstruction so it is corrected promptly under laboratory procedures. Credit emergency delay, possible worsening of injury, removal/reporting and prevention.
4. Goggles: PPE. Training: administrative control. Suitable containment cabinet: engineering control. Correct handwashing: safe work practice (often grouped with administrative controls). One mark each.

Source and coverage notes

SCI 101 lecture chapters 1-3: accident prevention, chemical hazards, laboratory techniques, emergency principles and biosafety. This is exam study material; real laboratory work follows the current instructor procedure and chemical-specific safety information.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.